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MLA0401

COS- Assessment

(1) A ware house robot :-

→ A ware house robot works with Dynamic programming

→ Dynamic programming helps the robot to find its own path. avoiding obstacles

for example:-

If a drone wants to deliver an product from starting point to ending point the drone should make its own decision for finding correct path, avoiding obstacles, delivering at correct place

The robot should know the exact cost for this action and how much battery should need.

(2) Monte carlo Contral :-

→ monte carlo control is nothing but to update the policy means how the agent or customer improves recommendations with experience

→ how this actually works means if a customer is spending time to buy a

product checking all the details it will get positive reward. If the customer didn't consider the product means it will get -ve reward

for example,

if you are having a company or business the customers wanted to continue the membership means you will get a positive reward. If the customers wanted to stop that membership means you will get a negative reward.

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(3) SARSA : ~~State~~.

→ it is nothing but State, Action, Reward again State Action

→ first the robot will select a state after that it will perform an action for that action it will be rewarded

$$r(S,a) = (S,a) + \alpha[r + \gamma Q(S' + a') - Q(S,a)]$$

→ by performing more actions it will get more +ve rewards and also it will improve its movements.

Robotic performance →

→ It is an

→ weather

action

all the details it will get
If the customer didn't
means it will get
product
business
the

Robotic performance → Reinforcement Algorithm

→ It is an reinforcement Algorithm
→ weather the robot is performing correct actions or not

for example:
If a drone is delivering orders to the customers. If the drone is delivery correctly or not, weather it is avoiding obstacles or not.

→ If it delivers correctly, then it will rewarded
→ If it didn't deliver correctly, then it won't be rewarded

result: by using trail and error methods the Robot will imbrove it moments and performance.

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(5) Q-learning:

→ Q-table
→ Simple environment
→ low scalability

DQN

→ Neural networks
→ Complex environment
→ high scalability

→ Simples roads

→ easier and smaller for small roads

→ Traffic, complex choicy

→ complex for Traffic roads.

(7)

A2C

synchronized learning

fast (speed)

more stable

efficient

stable

dependanty

A3C

Asynchronized learning

us of faster (casually)

less stable

uses multiple workers

fast learner.

Independent.